Econ 30065: Microeconomics
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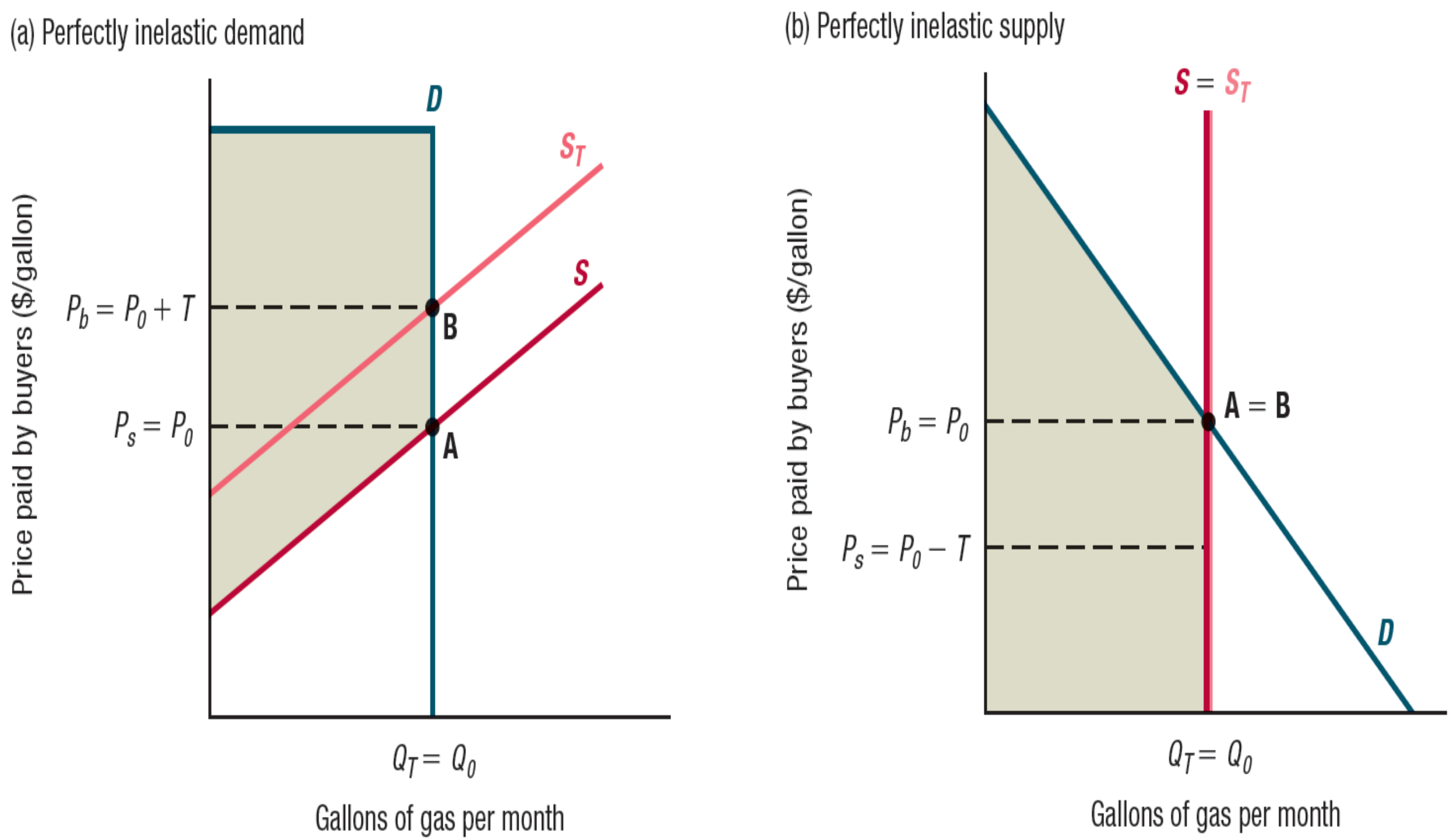
Market intervention II

Which goods should be taxed?

- Tax goods for which the deadweight loss of taxation will be low
- Deadweight loss of taxation will be low if either the demand or the supply curve is very inelastic:
  - If supply or demand is perfectly inelastic, for example, there is no deadweight loss:

    The tax does not change the quantity bought and sold

Taxation with No Deadweight Loss



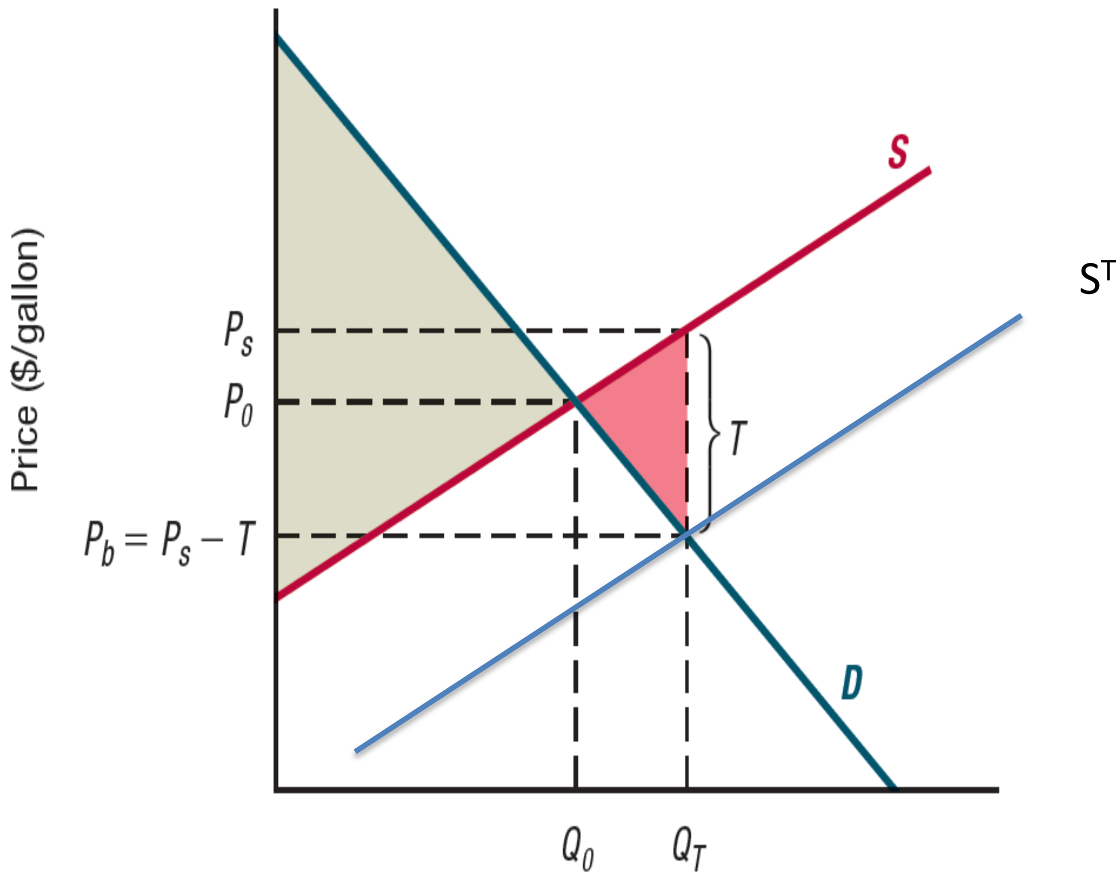
Which goods should be taxed?

- The good with less elastic demand or supply should face a larger tax
- Distributional considerations can also affect the choice of goods to tax
  - Many goods with inelastic demand are necessities (e.g. bread, gas)
- Can taxation ever *increase* Aggregate Surplus?
  - Yes, if the item that is taxed is a "bad" (e.g. pollution) (negative externalities)

Subsidies

- **Subsidy:** payment that reduces the amount that buyers pay for a good or increases the amount that sellers receive (example: subsidy of 3$ on each unit sold or bought)
- In competitive markets, subsidies create:
  - increase sales of the affected goods
  - **deadweight loss**

Effect of a subsidy T to firms



Effect of a subsidy T to firms

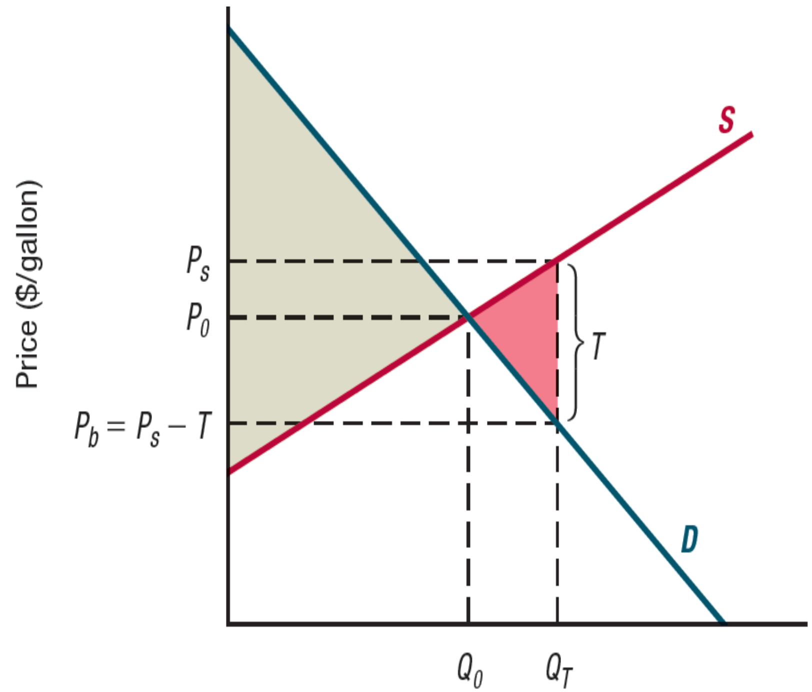
- For any price paid by consumers, firms now receive more than when there is no subsidy
  - willing to supply more than before
  - Supply curve with the subsidy is a distance T below the original supply curve
- New equilibrium price paid by consumers is price at which the demand curve and new supply curve cross
  - Amount bought and sold increases
  - Price paid by consumers falls; price received by firms increases
- In a competitive market the benefit of a subsidy is shared by consumers and firms

Effect of a subsidy on buyers

- Regardless of who receives the subsidy by the government, consumers and producers obtain the same benefit. (like in taxes)
- The share of the benefit received depends on the elasticity of the demand and supply curves

Welfare effects of a subsidy

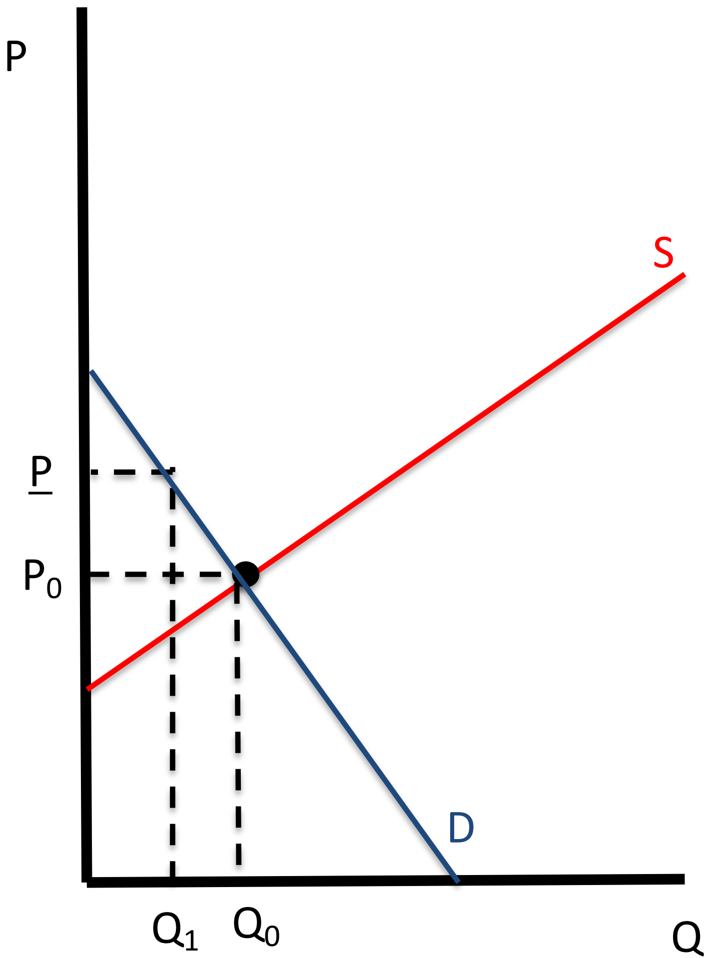
- Welfare analysis of a subsidy shows consumers and producers both gain
- The sum of the reduction in price to consumers and the increase in price to firms exactly equals the size of the subsidy: T
- Aggregate surplus falls (CS+PS+GR)
  - This is because the government incurs an expense, the per-unit subsidy times the number of units sold

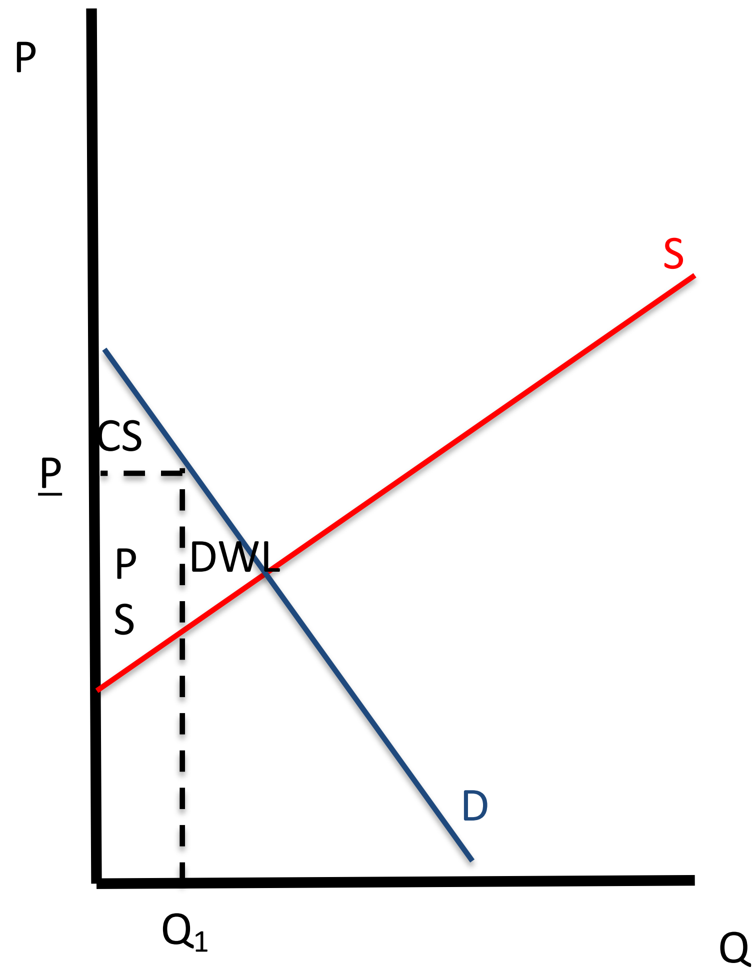


Policies to raise price

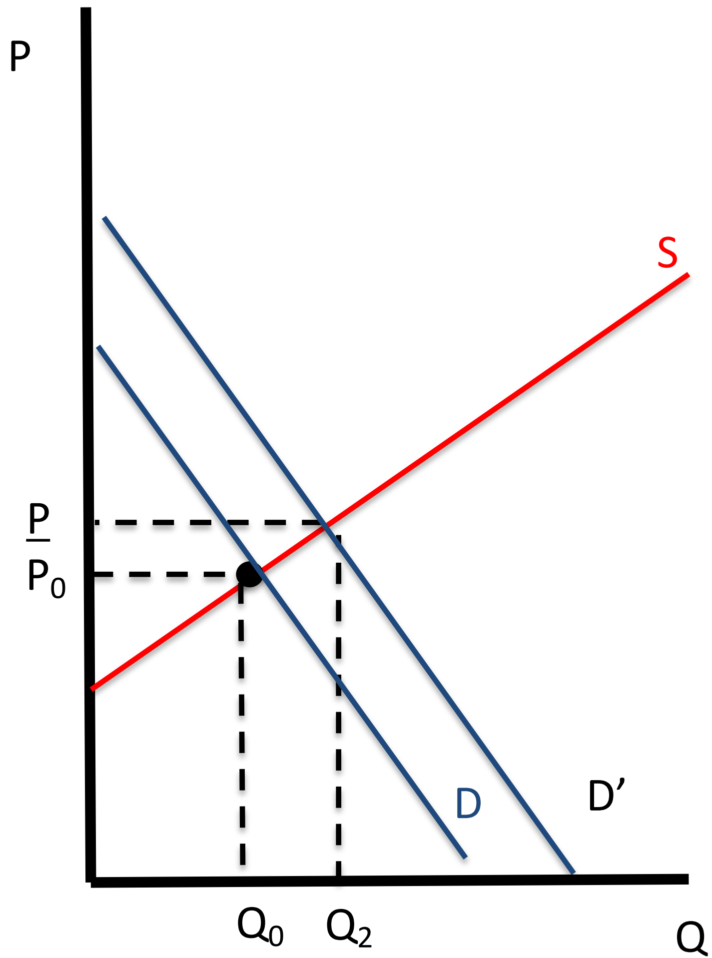
- Price floors
    - Minimum price that sellers can charge
    - Problem: monitoring
- Price support
    - Increase in market demand
    - overproduction
- Production quotas
    - Restriction of supply
    - Voluntary production reduction

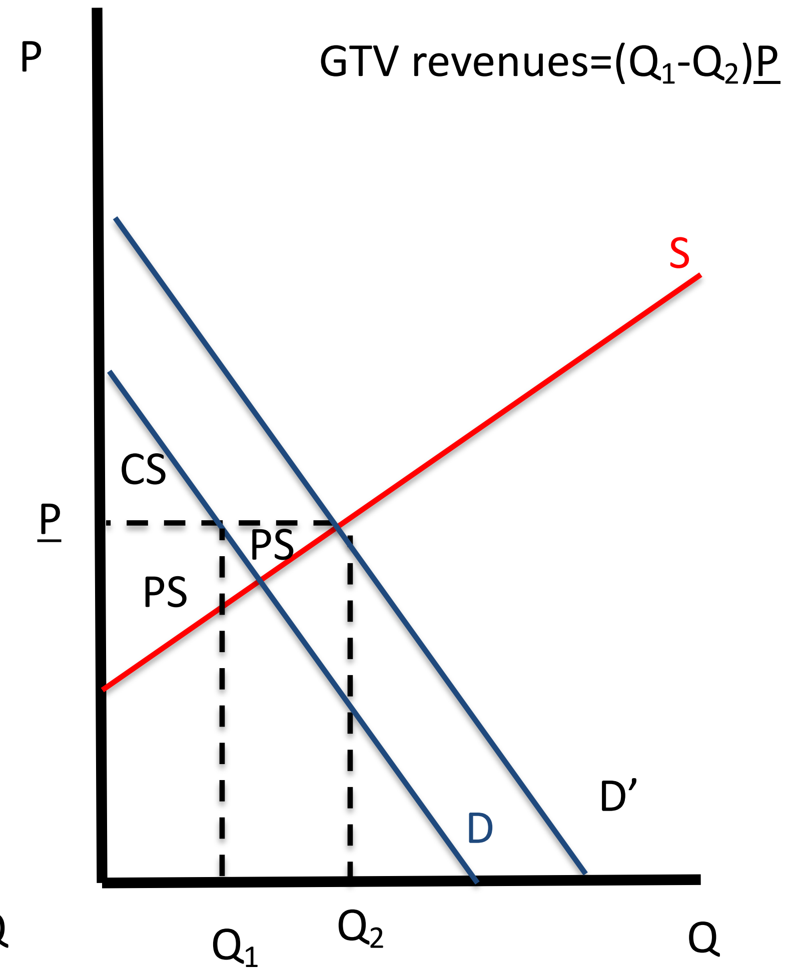
Price floor



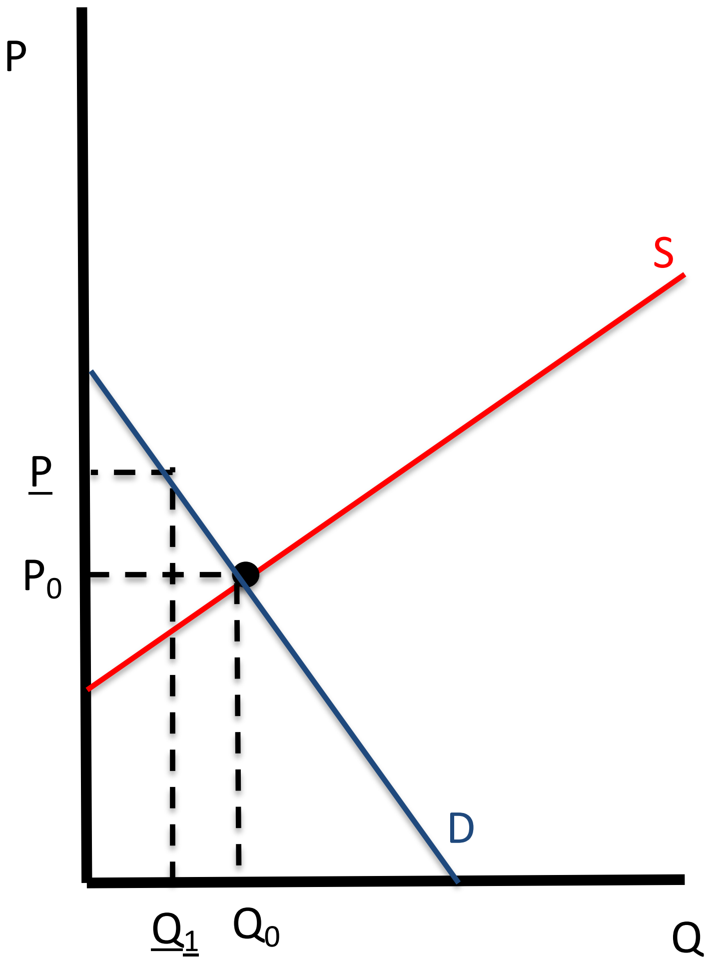


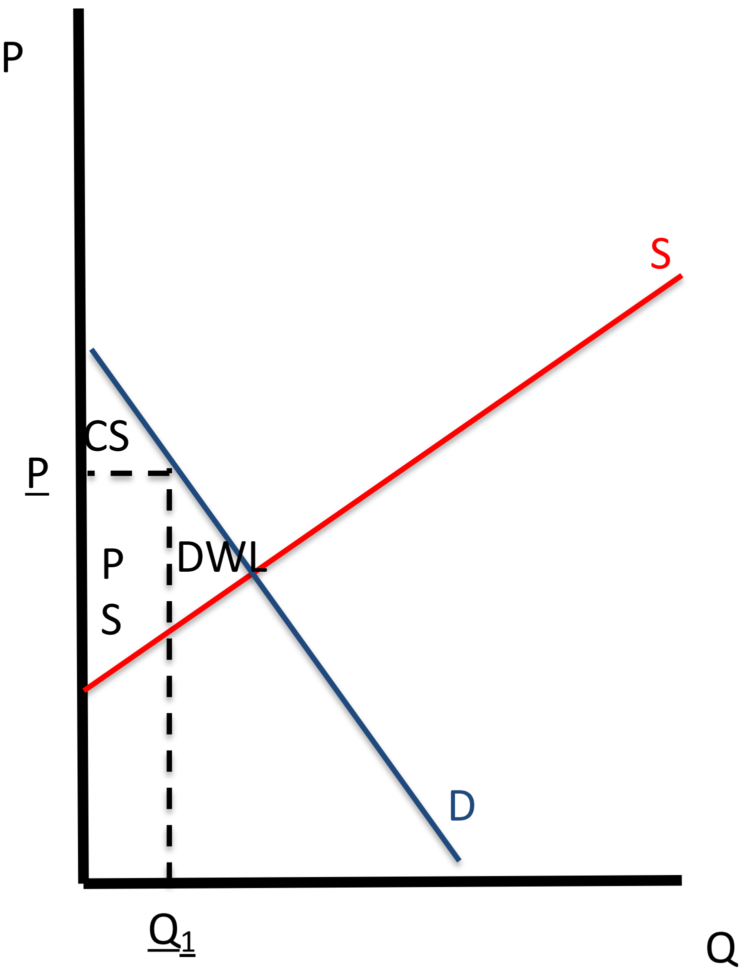
Price support





Production quota

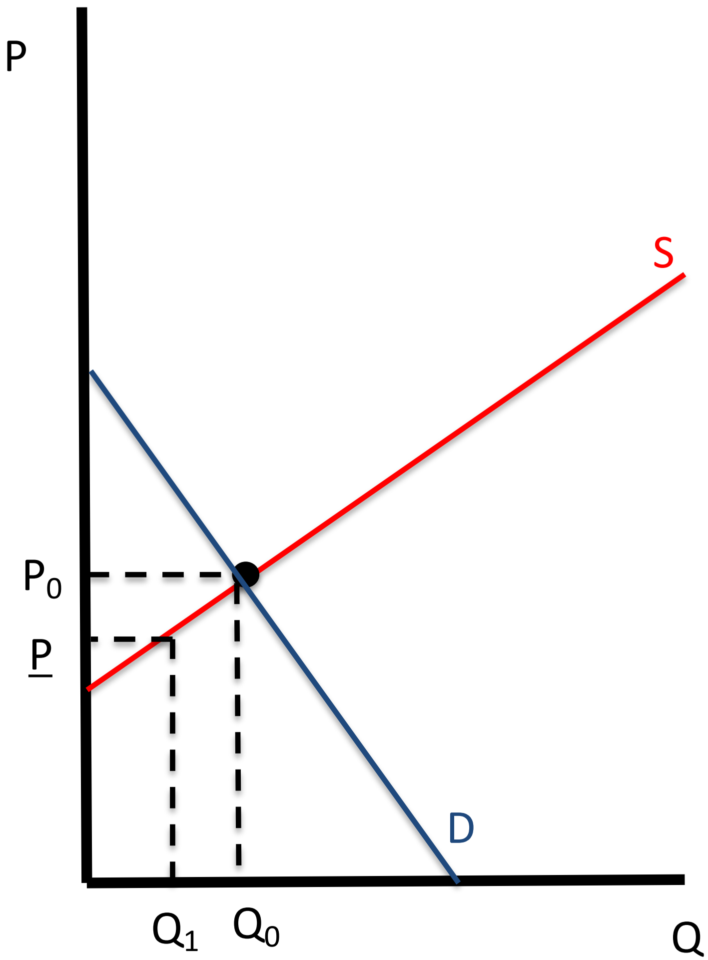


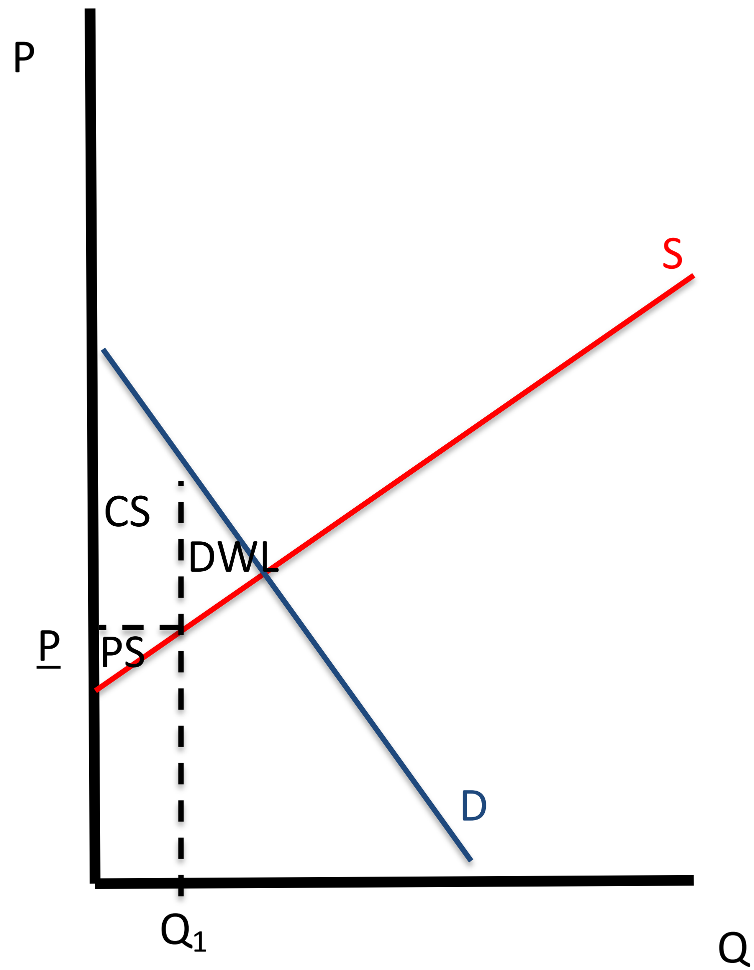


Policies to reduce price

- Price ceiling
  - Maximum price that sellers can charge

Price ceiling





Monopoly regulation

- Government regulates monopolies
  - prices closer to MC
  - Natural monopoly P=AC (second-best)
  - Government ownership